

Population CSV Format

This document describes the format for `population.csv` files used by the scorecard module (M8).

Column Headers

Column	Required	Description
admin_area_2	Optional	Second-level admin area (e.g., state, county)
admin_area_3	Optional	Third-level admin area (e.g., district, sub-county)
admin_area_4	Optional	Fourth-level admin area (e.g., ward, LGA)
year	Required	Four-digit year (e.g., 2023)
population_type	Required	Type of population count (see options below)
count	Required	Population count for this area/year/type

Notes:

- Include at least one admin area column that matches your HMIS data granularity
- The script uses the finest common admin level between population and HMIS data for joining
- Parent admin columns (e.g., `admin_area_2` when you have `admin_area_3`) are optional but can help with human readability

Population Types

ID	Description
total_population	Total population
u5	Under 5 population
u1	Under 1 population
wra	Women of reproductive age (15-49)
births	Expected births

ID	Description
pregnancies	Expected pregnancies

Only include the population types needed by your calculated indicators.

Example

```
admin_area_2,admin_area_3,year,population_type,count
"Northern Region","District A",2022,total_population,150000
"Northern Region","District A",2023,total_population,154500
"Northern Region","District A",2024,total_population,159135
"Northern Region","District A",2022,u1,4500
"Northern Region","District A",2023,u1,4635
"Northern Region","District A",2024,u1,4774
"Northern Region","District B",2022,total_population,200000
"Northern Region","District B",2023,total_population,206000
"Northern Region","District B",2024,total_population,212180
```

Interpolation

The script automatically interpolates between annual values to estimate monthly population:

- For months within available years: linear interpolation between adjacent years
- For months beyond available years: extrapolation using growth rate from nearest years
- Interpolation assumes annual values represent January 1 of each year

Legacy Format

Files using the old `period_id` column (e.g., `202301`) are still supported and will be automatically converted to the `year` format.